



Advanced Terraform on Azure

Lab 4 – Remote State & Multiple Environments

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Expected Duration

This lab should take 35-45 minutes to complete

Teaching Points

This lab expands your Module 3 configuration by introducing remote state and multi-environment structure. Local state files work only for a single user on a single machine. Remote state enables collaborative, safe Terraform usage for teams.

Key principles:

- terraform.tfvars holds global defaults shared across all environments.
- dev/test/prod var-files override only environment-specific differences.
- Backend state keys isolate environments.

You will:

- Build a storage account to hold Terraform state.
- Migrate your existing Module 3 state into remote storage.
- Create multiple environment configurations (dev, test, prod).

- Learn how Terraform uses state 'keys' to isolate environments.
- Understand why backend values must be literal, not variables.

Before You Begin

Ensure you are familiar with the outcome of Module 3. A copy of the final code revision has been carried forward into the lab4/original and lab4/guided folders. You will start by confirming that the Module 3 solution is functional before migrating to remote state.

- Work only inside the lab4/guided folder.
- Use the lab4/original folder for roll-back if needed
- Use --var-file=<env>.tfvars for environment-specific runs.
- Use terraform.tfvars for global shared values.
- Always specify --var-file explicitly when deploying.
- When switching backend state keys, always run Terraform with -reconfigure.
- Avoid editing the state-backend configuration after creation - it is intentionally minimal.

Try It Yourself (Optional)

Before you begin, think about the following:

- What happens if two people run terraform apply on the same local state file?
- How could a production environment be protected by forcing Terraform to store state remotely?
- How might three state files (dev/test/prod) live inside a single Azure storage container?

Phase 1 – Baseline: Run Starter Code Locally

1. In **lab4/guided**, update **providers.tf** with your **subscription id**.
2. A checker script is provided to identify the statefile currently in use. Run this file now ..

```
cd c:\qatfadavaz\lab4\guided
```

```
.\Print-Statefile.ps1
```

```
PS C:\qatfadavaz\lab4\guided> .\Print-Statefile.ps1

Terraform backend identity:
<backend cache not found>

This usually means:
- You have not run 'terraform init' in this folder yet, OR
- A local state file is in use (terraform.tfstate) rather than a remote backend.
```

3. Run ...

```
az login
terraform init
terraform apply --auto-approve
```

4. Re-run Print-Statefile.ps1 ...

```
PS C:\qatfadavaz\lab4\guided> .\Print-Statefile.ps1

Terraform backend identity:
<backend cache not found>

This usually means:
- You have not run 'terraform init' in this folder yet, OR
- A local state file is in use (terraform.tfstate) rather than a remote backend.

Note: A local state file exists here:
terraform.tfstate
```

5. Confirm that **.terraform.lock.hcl** and **terraform.tfstate** exist locally.
6. Use the Azure portal to confirm that resource group, VNet, NSG, and subnets creation succeeds.

You now have a working baseline deployment before migrating to remote state.

Phase 2 – Create the Remote State Backend

7. A course-wide **artifacts** repository exists at the root of your courseware folder. Navigate to the provided **lz-state-backend-bootstrap** folder in **artifacts**. This folder contains only the backend infrastructure code and is an example of a bootstrap configuration.
8. Examine the provided files;
- providers.tf - Azure provider
 - variables.tf - subscription + RG name variables
 - main.tf - resource group, storage account, container
 - outputs.tf - expose RG name, SA name, container name
 - terraform.tfvars - subscription and naming values
9. In **artifacts\lz-state-backend-bootstrap**, update **terraform.tfvars** with your **subscription id** and a random **suffix** for your storage account, which must be uniquely named.

Important !

This remote backend will remain in place for the remainder of this course. Future labs assume its existence and configuration as detailed above. Change **ONLY** your subscription ID and random suffix. Make a note of the suffix chosen as you will require it for future labs. Should the entire lab environment be reset then ensure you recreate this backend first before re-attempting any lab 5 onwards.

10. Run ...

```
cd c:\qatfadavaz\artifacts\lz-state-backend-bootstrap
terraform init
terraform apply --auto-approve
```

In the Azure Portal, verify that the storage account and container are created.

This creates the remote backend prerequisites.

Phase 3 – Configure and Migrate to Remote Backend

11. Return to the **lab4\guided** folder and open **providers.tf**.
12. Add this backend block inside terraform {}, updating the storage account name placeholder with your suffix:

```
backend "azurerm" {  
  resource_group_name = "landing-zone-shared-tfstate-rg"  
  storage_account_name = "advtfstate<your suffix>"  
  container_name      = "tfstate"  
  key = "default.tfstate"  
}
```

Note: Backend blocks cannot use variables or expressions. Values **MUST** be literal.

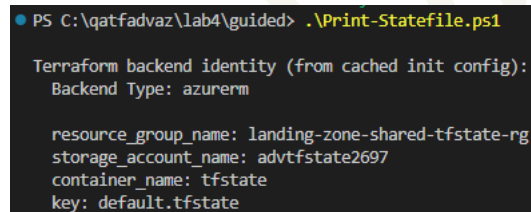
13. Run ...

```
cd c:\qatfadvaz\lab4\guided  
terraform init -migrate-state
```

Enter 'yes' when prompted to migrate

Terraform moves your local terraform.tfstate into the Azure storage container and clears your local terraform.tfstate file.

14. Re-run Print-Statefile.ps1 ...



```
PS C:\qatfadvaz\lab4\guided> .\Print-Statefile.ps1  
  
Terraform backend identity (from cached init config):  
Backend Type: azurerm  
  
resource_group_name: landing-zone-shared-tfstate-rg  
storage_account_name: advtfstate2697  
container_name: tfstate  
key: default.tfstate
```

15. Review the **terraform.tfstate** file in .terraform and note how it references the container name, key, resource group and storage account...

```

"container_name": "tfstate",
"endpoint": null,
"environment": null,
"key": "default.tfstate",
"lookup_blob_endpoint": null,
"metadata_host": null,
"msi_endpoint": null,
"oidc_request_token": null,
"oidc_request_url": null,
"oidc_token": null,
"oidc_token_file_path": null,
"resource_group_name": "landing-zone-shared-tfstate-rg",
"sas_token": null,
"snapshot": null,
"storage_account_name": "advtfstate2697",
"subscription_id": null,

```

You now have a remote, team-safe, backend state. It is using **default.tfstate** and would suffice if a single environment were being maintained.

Phase 4 – Introduce Multiple Environments (DEV, TEST, PROD)

16. For clarity, we will destroy the default deployment ahead of moving to a multi-environment configuration that will create separate state files per environment.

17. Run ...

terraform destroy --auto-approve

18. Each environment requires a named tfvars file...

- Rename terraform.tfvars to **dev.tfvars**
- Duplicate dev.tfvars as **test.tfvars**
- Duplicate dev.tfvars as **prod.tfvars**

19. Update **test.tfvars** ...

```

env = "test"
vnet_address_space = "10.1.0.0/16"

subnet_cidrs ...
  data = "10.1.2.0/24"
  app = "10.1.1.0/24"
  endpoints = "10.1.3.0/24"

```

20. Update **prod.tfvars** ...

```

env = "prod"
vnet_address_space = "10.2.0.0/16"

subnet_cidrs ...
  data = "10.2.2.0/24"

```

```
app = "10.2.1.0/24"
endpoints = "10.2.3.0/24"
```

1. For each environment, reconfigure the backend to use a different state key:

```
terraform init -reconfigure -backend-config="key=dev.tfstate"
terraform apply --var-file=dev.tfvars --auto-approve
```

Check that the local terraform.tfstate file in .terraform is updated to reflect the new key
Run **Print-Statefile.ps1** to confirm

```
terraform init -reconfigure -backend-config="key=test.tfstate"
terraform apply --var-file=test.tfvars --auto-approve
```

Repeat the statefile checks

```
terraform init -reconfigure -backend-config="key=prod.tfstate"
terraform apply --var-file=prod.tfvars --auto-approve
```

Repeat the statefile checks

2. Verify in Azure:

- Three Resource groups exist
- Three VNets exist
- Three NSGs exist
- Each environment has its own three subnets
- All NSGs use the same rule

3. Verify in Azure Storage that alongside the now redundant default.tfstate file there are environment specific state files...

- dev.tfstate
- test.tfstate
- prod.tfstate

Takeaway

You now have a multi-environment Terraform pattern.

Terraform always stores backend *configuration* locally in .terraform/terraform.tfstate, so always reconfigure when switching. To prevent accidental cross-environment drifts, follow this sequence when switching between environments:

```
terraform init -reconfigure -backend-config="key=..."
terraform plan/apply/destroy -var-file=...
```

Phase 5 – Clean Up

4. Run ...

```
cd c:\qatfadvaz\lab4\guided
terraform init -reconfigure -backend-config="key=dev.tfstate"
terraform destroy --var-file=dev.tfvars --auto-approve
terraform init -reconfigure -backend-config="key=test.tfstate"
terraform destroy --var-file=test.tfvars --auto-approve
terraform init -reconfigure -backend-config="key=prod.tfstate"
terraform destroy --var-file=prod.tfvars --auto-approve
```

Use the portal to verify all lab resources other than the remote backend have been destroyed.

***** Congratulations, you have completed this lab. *****

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